May 2026

AI-Driven Tactical Intelligence in Football

Tnt present key

Graph Neural Networks, Computer Vision, Transformers & Multimodal AI



PhD Research Presentation

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May 2026

Football as a Complex AI Problem

- High-dimensionality state space
- Multi-agent interactions
- Partial observability
- Sparse, long-horizon rewards
- Continuous action control
- Non-stationary environments



Multi-Agent Systems

Spatial Reasoning

Temporal Modeling

Sequential Decisions

Context Understanding

Core Research Problem & Existing Limitations

Current Analytics Focus

Event-based tracking

Static tactic models

Simple prediction trends

Why Insufficient?

Ignores agent dynamics

Lacks context-awareness

Poor reward handling

Research Gap

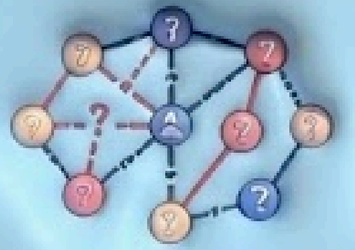
Current Limits Data Insight

Temporal Complexity

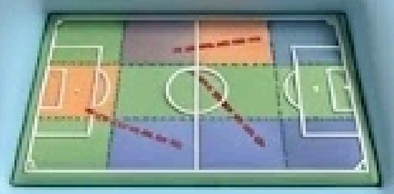
Current Limits



- Custom full-team 3D positional group
- Isolated player
- Isolated player metrics

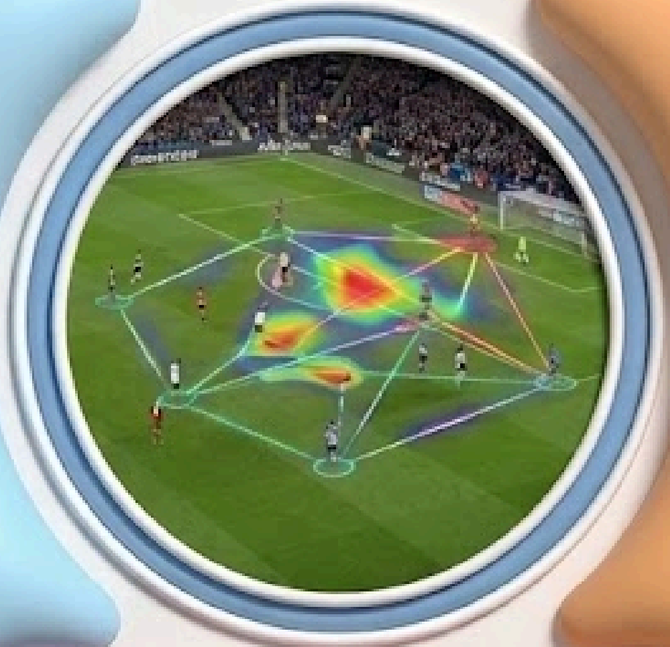


- Collective context network diagram

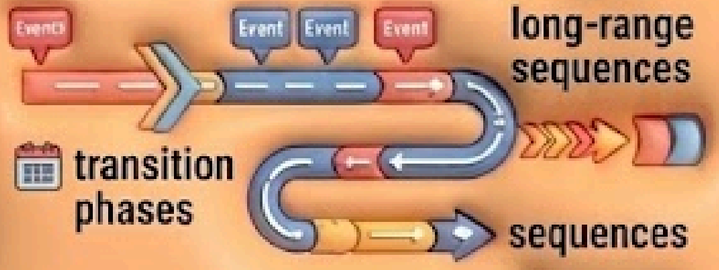


- Multiple pitch areas

missed connections



The Challenge

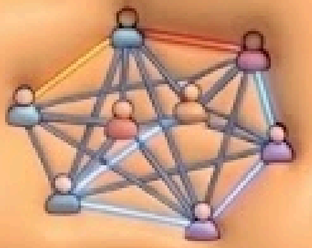


long-range sequences

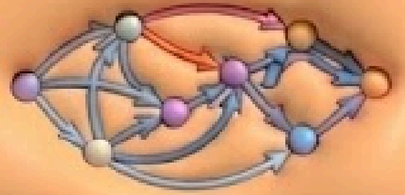
transition phases

sequences

Dynamic dependencies between ten players



Predictive 3D future tactical patterns



Fundamental Gap
Lack of a Unified Brain for Football Tactics

Team GNN Brain

Spatio-Temporal Fusion

Sequential Event Brain

Match Contextual Intelligence

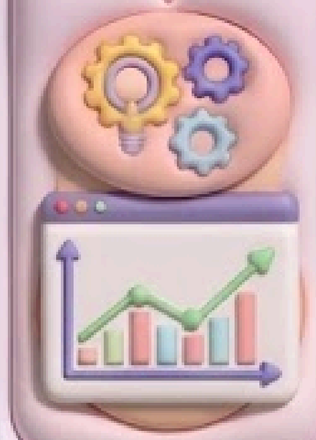
Tactical Pattern Recognition

Panoramic Tactical Structure Modeling with Graph Neural Networks

GNNs capture complex, dynamic player and team interactions that traditional, static methods cannot analyze.







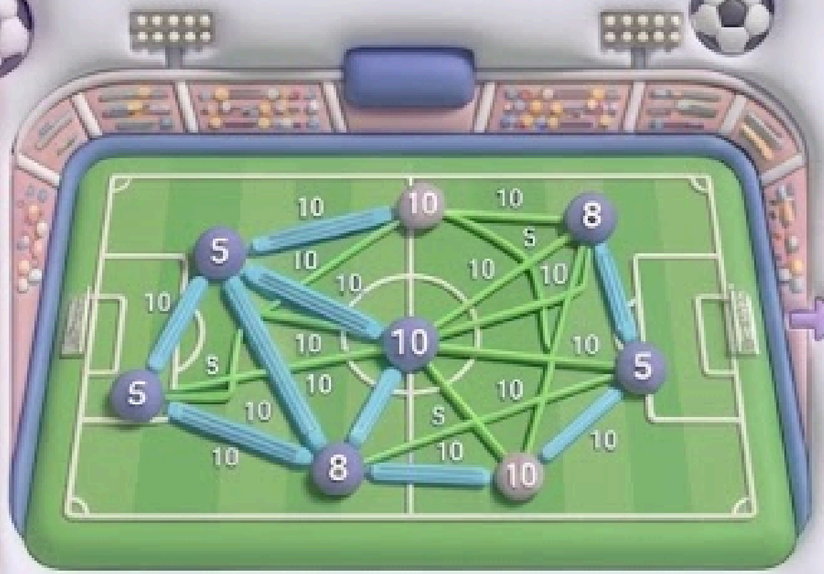
GNN Idea

PLAYERS AS NODES
Each player is a graph node with unique features (speed, position, stete).

INTERACTIONS & PASSES AS EDGES
Dynamic interactions and ball passes define connections.

LEARN DYNAMIC PATTERNS
GNNs train on sequences of these interaction graphs to understand team tactics.

Passing Network Example



GNN Pipeline

INPUT: GLOBAL PLAYER GRAPHS (GRAPH)

GNN PROCESSING & LAYERED CONVOLUTION



OUTPUT: TACTICAL PREDICTIONS

- PRESSING
- THREATS
- LINE-BREAKING...

Applications

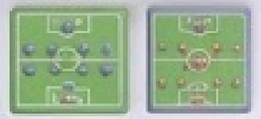
1. LINE-BREAKING PASS PREDICTION



2. PRESSING DETECTION



3. TEAM SHAPE ANALYSIS



4. PLAYER INFLUENCE ESTIMATION



5. OPPONENT RISK ASSESSMENT



Mathematical Formulation

GRAPH $G = (V, E)$

GRAPH CONVOLUTION

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right)$$

- A is the adjacency matrix with self-loops.
- D is the diagonal degree matrix for normalization.
- H represents player node features.
- W is the trainable weight matrix.

WHERE $\tilde{A} = A + I$

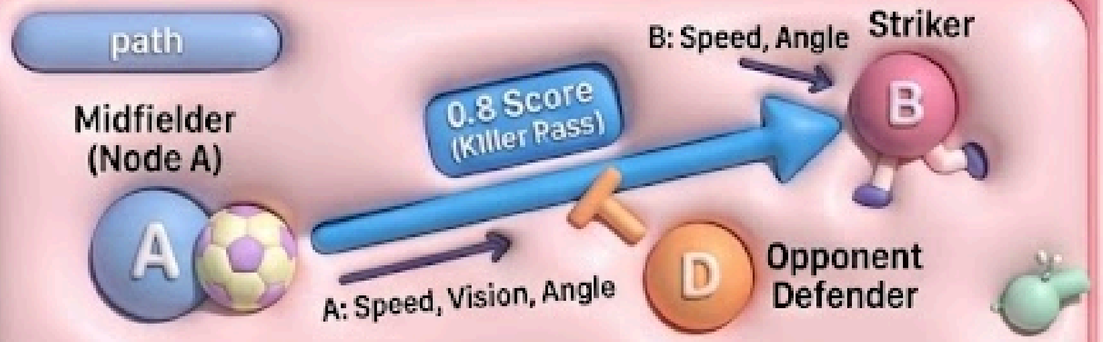
- $\tilde{D}$ is the diagonal degree matrix of $\tilde{A}$, used for normalization.

Applications

- Line-Breaking Pass Prediction
- Pressing Detection
- Team Shape Analysis
- Player Influence Estimation

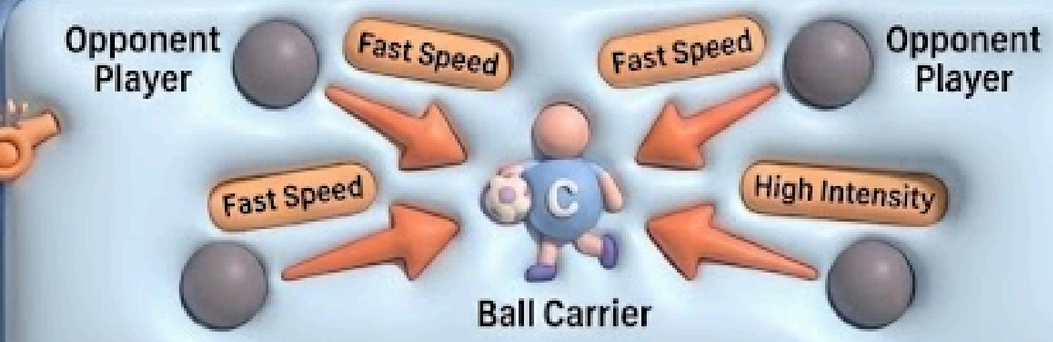
TACTICAL GNN PREDICTIONS

1. Line-breaking Pass Prediction (Link Prediction)



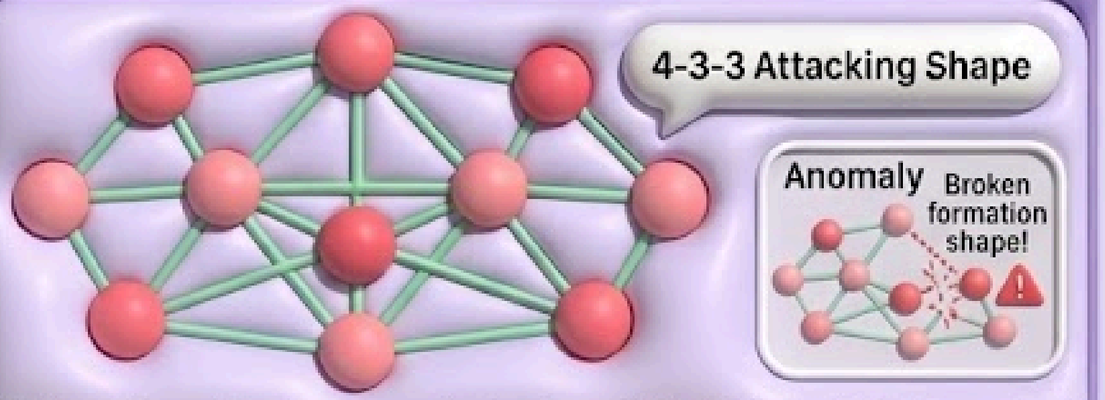
Logic: Midfielder (A) to Striker (B), factoring "Space".
defender (D) poses negative influence. **Prediction Score: High!**

2. Pressing Detection (Node Classification)



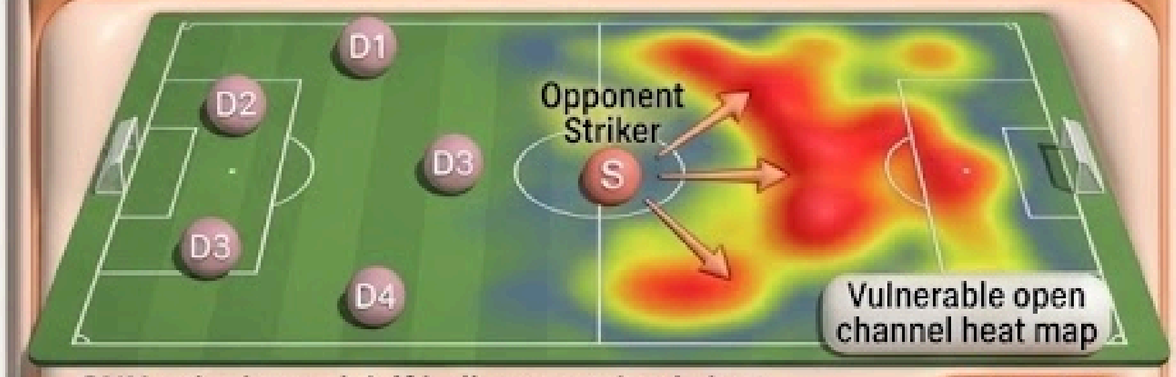
GNN analyzes 3-4 Opponents converging. Nodes labeled "Pressing!"
Output: High Pressing Detected from Left Wing!

3. Team Shape Analysis (Graph Classification)



GNN analyzes 11-player coordinates for a Global Vector (hG) "Snapshot".
Compares to memory of known formations. Detects breaks.

4. Counter-Attack Vulnerability (Graph Regression)



GNN calculates risk if ball possession is lost.
High dispersion of defense nodes and open channels.
Regression Score: 85% High Risk! Alerts coach: Defensive line too thin.

Defensive
thin

Problem 2: Computer Vision for Tactical Understanding

Extract tactical information from broadcast videos without expensive tracking systems.

DATA PRE-PROCESSING

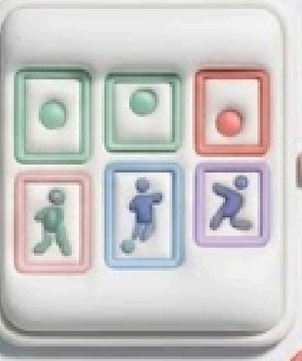
- Broadcast Video Denoising
- Camera Calibration (Bird's-Eye Projection)
- Homography Mapping to Pitch Coordinates

TACTICAL METRICS EXTRACTION

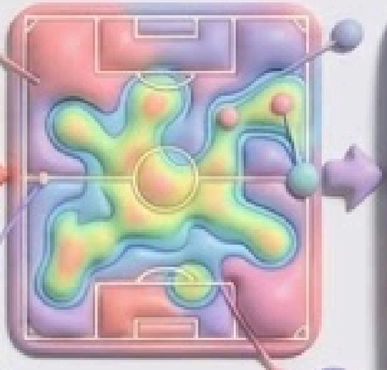
- Player Velocity & Acceleration
- Inter-player Dispersions
- Team Centroid Mapping

Broadcast Video → Player Detection (YOLOv8) → Tracking (ByteTrack) → Feature Extraction (Positional Map) → Tactical Phase Classification









- start Build-up
- Pressing
- Transition
- Attack

MODEL EVALUATION

- Detection mAP Score (Mean Average Precision)
- Tracking Accuracy (MOTA/MOTP)
- Phase Classification F1-Score

FURTHER APPLICATIONS

- Opponent Team Profiling
- Formation Anomaly Detection
- Expected Goals (xG) Calculation Support

Detection Loss

$$L = L_{box} + L_{cls} + L_{obj}$$

Temporal Modeling (LSTM)

$$h_t = f(Wx_t + Uh_{t-1} + b)$$

LSTM LSTM

Tactical Output (Heatmap)



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Part 2: Extended Tactical AI Applications

Extract tactical information for and **Positional Heatmap**

DATA PRE-PROCESSING

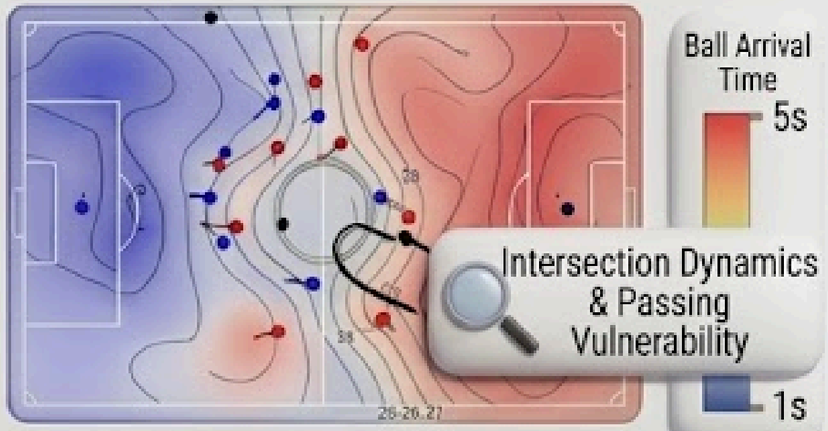
- Broadcast Video Denoising
- Camera Calibration (Bird's Eye Projection)
- Homography Mapping to Pitch Coordinates

A. Pitch Control: Modeling (expands from Positional Map)

Input:

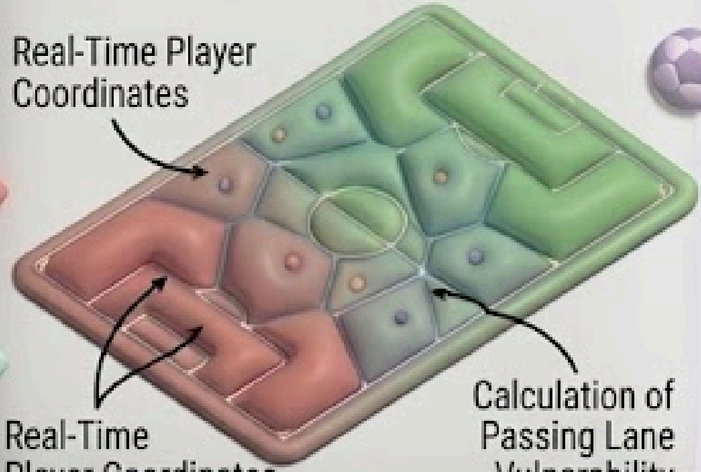
time motion

Output:
Dynamic Dynamic Player Positions & Physics Models



Contour Map with Key

B. Dynamic Voronoi Diagrams



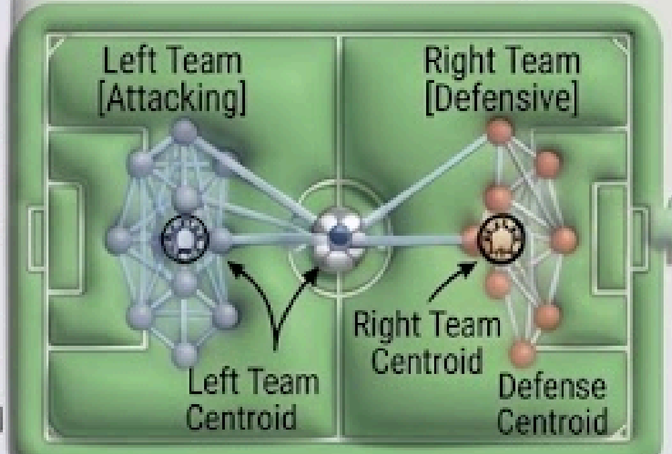
MODEL EVALUATION

- Detection MAP Score (Mean Average Precision)
- Tracking Accuracy (MOTA/MOTP)
- Phase Classification F1-Score

TEAM CENTROID & DISPERSION

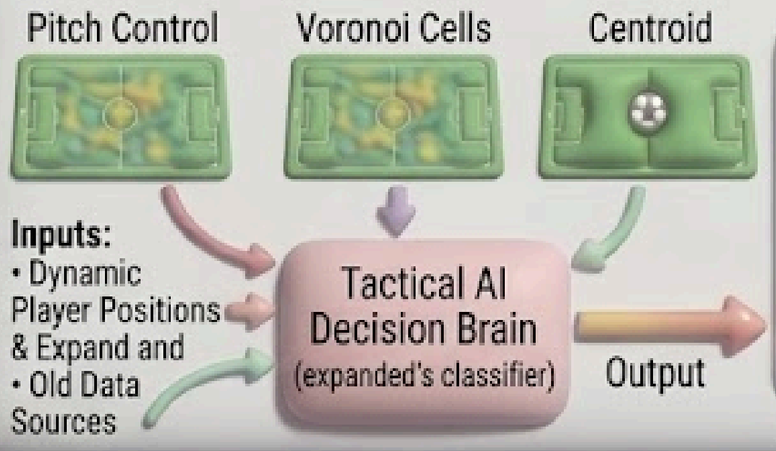
- Geometric Centroid [Team Center of Mass]
- Dispersion Halo [Inter-Player Spacing Network]
- Team Structural Compactness [GNN Structural Measure]

C. Team Centroid & Dispersion Analytics





Integrated Tactical Understanding



- Context-Aware Tactical Modeling
- Optimized Play Prediction

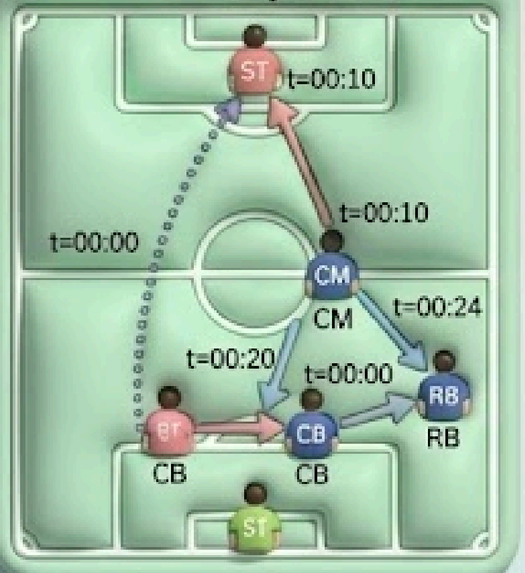
Advanced Tactical Insights

Next Slide: Advanced Modeling

Problem 3 (Part 1): Transformer-Based Tactical Sequence Intelligence

Tactics unfold over time. Understanding requires modeling long-range dependencies in sequential events.

Football Match Event View (Spatial)



Event Sequence Example

00:00	CB → RB
00:02	RB → CM
00:04	CM → LW
00:06	Dribble
00:08	LW → ST
00:10	ST

...

Transformer Architecture

Input Embedding (Spatial + Temporal)

↓

Positional Encoding

↓

$+t$

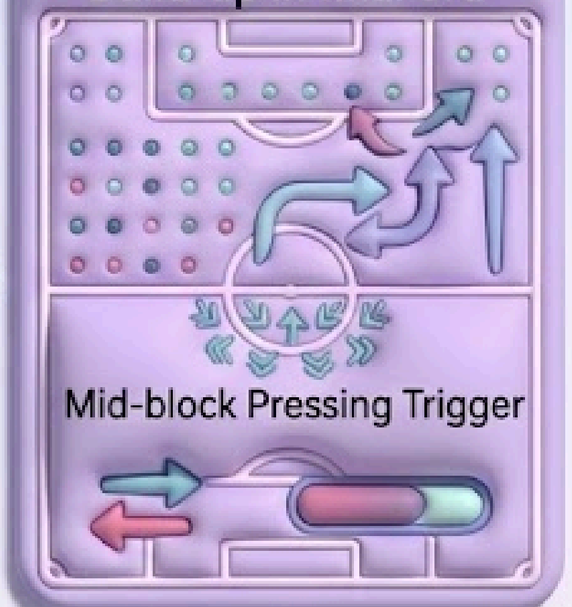
↓

Transformer Encoder (Self-Attention Layers)

↓

Tactical Representation

Build-up in final 3rd



Tactical Pattern Output (Contextual)

Self-Attention Mechanism

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

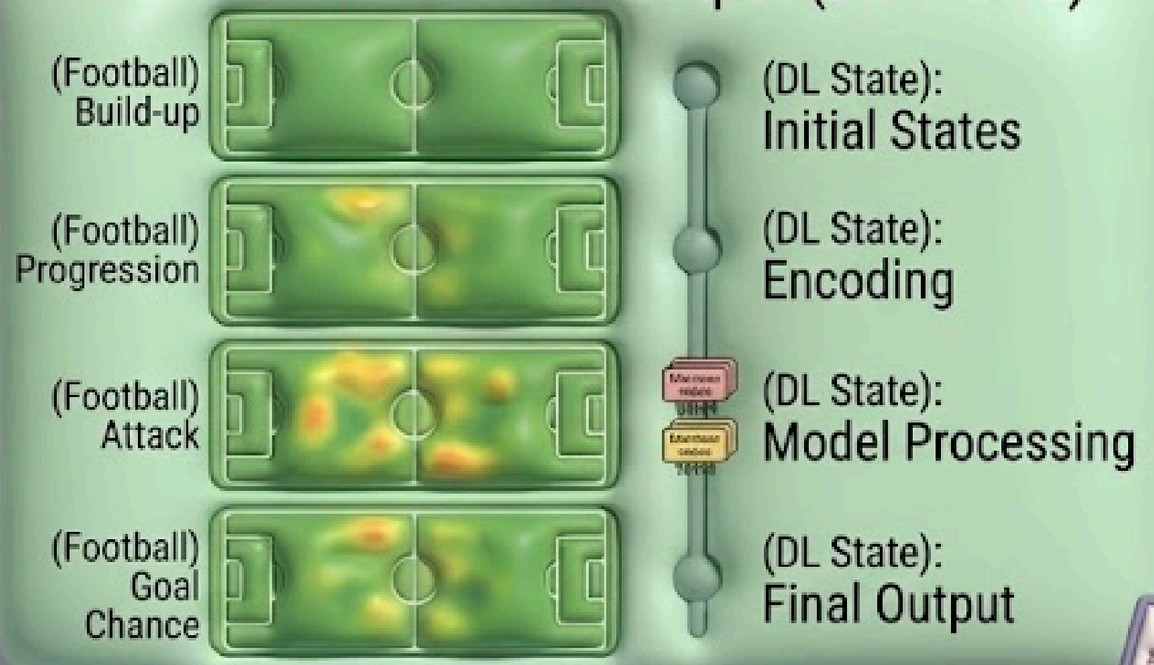
Problem 3 (Part 2): Transformer-Based Tactical Sequence Intelligence

Capture Global Context Across an Entire Play

Self-Attention (Q-K Interaction)



Transformer Tactical Output (DL States)



AI capabilities -> Football Impact

DL	Detect Long-range Dependencies	Handle Entire Match Sequences	Improve Context Understanding
	Earlier pass affects later shot	Earlier pass affects later shot	Ceconizes tactical understanding
			Analyze Event + Positional Data
			Combines loccations data, tinal

Unified Multimodal Tactical Intelligence



VIDEO DATA

Capture visual info (player positions, ball trails, and movements)



MATCH EVENTS

Capture discrete events (specific tactical actions and outcomes)



TEAM GRAPHS

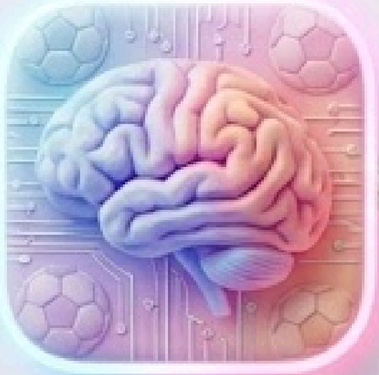
Capture interaction structures (passing network, proximity map, team shape)



PITCH GEOGRAPHY

Capture spatial coordinates and pitch zones (heatmap context)

Multimodal Transformer



$$F = f(V, E, G, P)$$

where **V** = video, **E** = event, **G** = graph, **P** = positions

HOLISTIC CONTEXT



Deeper insights: predictive lane analysis and goal probability maps from unified representation

ROBUSTNESS



Robustness: missing tactical data imputed to complete complex patterns

Research Vision



Develop next-generation AI systems that can:

- Understand football tactics at scale
- Provide real-time tactical insights
- Predict outcomes & tactical events
- Explain decisions & patterns
- Assist coaches, analysts & clubs
- Advance the future of sports AI

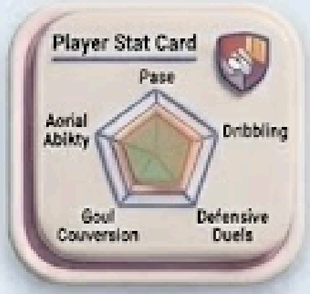
Impact & Applications



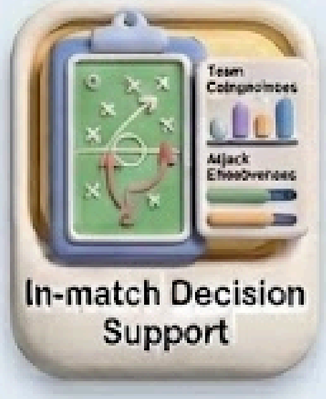
Tactical Analysis & Insights



Opponent Scouting & Match Prep



Player Evaluation & Recruitment



In-match Decision Support



AI Coaching Assistants


Thank You!
VAR
MVP
VAR